

Def. A linear operator  $E: V \rightarrow V$  is called a projection if  $E^2 = E$ .

Lemma. Let  $E: V \rightarrow V$  be a projection. Then,

1)  $\text{Im } E = \{x \in V \mid E(x) = x\}$ ,

2)  $V = \text{Im } E \oplus \ker E$

Proof.  $x \in \text{Im } E \Leftrightarrow x = E(\alpha)$  for some  $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$ . The converse is trivial!

Given  $x \in V$ ,  $x = E(x) + (x - E(x))$ . Clearly  $E(x) \in \text{Im } E$  and

$$E(x - E(x)) = E(x) - E^2(x) = (E - E^2)(x) = 0, \text{ so } x - E(x) \in \ker E.$$

And,  $E(y) = y$  and  $E(y) = 0$ , then  $y = 0$ . Thus  $\text{Im } E \cap \ker E = \{0\}$ .

Thm. Let  $V = W_1 \oplus W_2$ . Then, there is one and only one projection  $E: V \rightarrow V$  s.t.

$$\text{Im } E = W_1 \text{ and } \ker E = W_2$$

Proof. Let  $E$  be a projection. For  $\beta_1$ , basis for  $\text{Im } E$ , and  $\beta_2$ , basis for  $\ker E$ ,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r = \text{rank } E.$$

Comment: In General,

let  $\{\alpha_1, \dots, \alpha_k\}$  be a basis for  $\text{Im } E$  and  $\{\alpha_{k+1}, \dots, \alpha_n\}$  be a basis for  $\ker E$ .

Then, the invertible matrix  $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \end{bmatrix}$  satisfies

$$EA = [E\alpha_1 \dots E\alpha_n] = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix}$$

Therefore,  $E = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$  is we desired.

Resolution of identity.

Def. Given a vector space  $V$ , a collection  $\{E_i\}_{i=1}^k$  of linear operators on  $V$  satisfying

$$\underbrace{E_i^2 = E_i \text{ for each } 1 \leq i \leq k}_{\text{idempotent}}, \quad \underbrace{E_i \circ E_j = 0 \text{ if } i \neq j}_{\text{orthogonal}}, \quad \underbrace{I = E_1 + \dots + E_k}_{\text{complete}}.$$

is called the resolution of the identity.

In fact, the orthogonal with complete implies the idempotent, because

$$I = E_1 + \dots + E_k \Rightarrow E_i = E_i I = E_i E_1 + \dots + E_i E_k \Rightarrow E_i = E_i^2.$$

Moreover, if  $\{E_i\}_{i=1}^k$  is a resolution of the identity, then

$$V = \text{Im } E_1 \oplus \dots \oplus \text{Im } E_k$$

because: given  $\alpha \in V$ ,

$$\alpha = E_1 \alpha + \dots + E_k \alpha$$

so  $V = \text{Im } E_1 + \dots + \text{Im } E_k$ , that is, the complete gives the sum decomposition, and,

for each  $1 \leq i \leq k$ ,  $\alpha_i \in \text{Im } E_i$  be given. Suppose that  $\alpha_1 + \dots + \alpha_k = 0$ , then

$\alpha_i = E_i \beta_i$  for some  $\beta_i \in V$ , so, for each  $1 \leq j \leq k$ ,  $\underbrace{\text{orthogonal}}_{\text{idempotent}}$

$$0 = E_j \cdot 0 = E_j \alpha_1 + \dots + E_j \alpha_k = E_j E_1 \beta_1 + \dots + E_j E_k \beta_k = E_j E_1 \beta_1 + \dots + E_j E_j \beta_j = E_j E_j \beta_j = E_j \beta_j = \alpha_j.$$

Thus  $\text{Im } E_i$  are independent, so proved.

Lemma. Let  $T$  be a linear operator on  $V$ , and  $E_1, \dots, E_k$  be the projections satisfying

$$E_i \circ E_j = 0 \text{ if } i \neq j \text{ and } I = \sum_{i=1}^k E_i.$$

Then,

For each  $\text{Im } E_i$  is invariant under  $T$  if and only if  $TE_i = E_i T$  for each  $1 \leq i \leq k$ .

Proof. Suppose that  $\text{Im } E_1, \dots, \text{Im } E_k$  are invariant under  $T$ .

Given  $\alpha \in V$ ,  $\alpha = E_1 \alpha + \dots + E_k \alpha$  and so  $T\alpha = TE_1 \alpha + \dots + TE_k \alpha$ .

By the assumption, each  $TE_i \alpha \in \text{Im } E_i$ , so  $TE_i \alpha = E_i \beta_i$  for some  $\beta_i \in V$ .

Therefore,

$$E_i T \alpha = E_i (TE_1 \alpha + \dots + TE_k \alpha) = E_i (E_1 \beta_1 + \dots + E_k \beta_k) = E_i E_i \beta_i = E_i \beta_i = TE_i \alpha.$$

So converse. Conversely, let  $\alpha \in \text{Im } E_i$ . Then,  $T\alpha = TE_i \alpha = E_i T \alpha \in \text{Im } E_i$ , so proved.  $\square$

Thm. Let  $T$  be a diagonalizable linear operator on a finite-dimensional space  $V$ , with distinct eigenvalues  $c_1, \dots, c_k \in F$ .

Then, there exists a resolution of identity  $\{E_i\}_{i=1}^k$  on  $V$  satisfying

$$T = c_1 E_1 + \dots + c_k E_k \quad \text{and for each } 1 \leq i \leq k, \operatorname{Im} E_i = \ker(T - c_i I).$$

Proof. By the equivalent condition of diagonalizable, we obtain directly that

$$V = \ker(T - c_1 I) \oplus \dots \oplus \ker(T - c_k I)$$

Now, for each  $1 \leq i \leq k$ , let  $E_i$  be the projection on  $\ker(T - c_i I)$  along  $\bigoplus_{j \neq i} \ker(T - c_j I)$ .

Polynomial evaluation of linear operation.

Let  $T$  be a diagonalizable operation on a fin. dim space  $V$  with a resolution

$$T = c_1 E_1 + \dots + c_k E_k.$$

Then, the orthogonality gives that: for any polynomial  $g(t) \in F[t]$ ,

$$g(T) = g(c_1) E_1 + \dots + g(c_k) E_k.$$

So, we can use the Lagrange Polynomial,

$$p_j(t) = \prod_{i \neq j} \frac{(t - c_i)}{(c_j - c_i)}, \text{ so that } p_j(T) = E_j.$$

Thm. Let  $V$  be a fin. dim. v.s. over  $F$ .

If  $V = \bigoplus_{i=1}^k W_i$ , then there exist  $k$  linear operators  $E_i$  on  $V$  s.t.

$$E_i \circ E_j = \delta_{ij} \cdot E_i, \quad I = \sum_{i=1}^k E_i, \quad W_i = \text{Im } E_i \quad (1 \leq i \leq k) \quad \dots (*)$$

Conversely, if  $E_1, \dots, E_k$  are  $k$  linear operators on  $V$  which satisfy  $(*)$ , then

$$V = \bigoplus_{i=1}^k W_i$$

Proof. Suppose  $V = \bigoplus_{i=1}^k W_i$ . For each  $1 \leq i \leq k$ , define  $E_i: V \rightarrow V: \sum_{j=1}^k \alpha_j v_j \mapsto \alpha_i v_i$ . Proved.

Conversely, suppose that  $E_1, \dots, E_k$  are linear operators on  $V$  satisfying  $(*)$ .

Given  $\alpha \in V$ ,  $\alpha = E_1(\alpha) + \dots + E_k(\alpha) \in W_1 + \dots + W_k$ . And, to show uniqueness such expression,

let  $\alpha = \alpha_1 + \dots + \alpha_n$ , where  $\alpha_i \in W_i$ . And, let  $\alpha_i = E_i(\beta_i)$ , for each  $1 \leq i \leq k$ .

Now, for each  $j$ ,  $1 \leq j \leq k$ ,

$$E_j(\alpha) = \sum_{i=1}^k E_j(E_i(\alpha_i)) = \sum_{i=1}^k (E_j \circ E_i)(\alpha_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

Layer 2

Lemma. Let  $T$  be a linear operator on  $V$ , and  $E_1, \dots, E_k$  be the projections satisfying

$$E_i \circ E_j = 0 \text{ if } i \neq j \text{ and } I = \sum_{i=1}^k E_i.$$

Then,

For each  $\text{Im } E_i$  is invariant under  $T$  if and only if  $TE_i = E_i T$  for each  $1 \leq i \leq k$ .

Proof. Suppose that  $\text{Im } E_1, \dots, \text{Im } E_k$  are invariant under  $T$ .

Given  $\alpha \in V$ ,  $\alpha = E_1 \alpha + \dots + E_k \alpha$  and so  $T\alpha = TE_1 \alpha + \dots + TE_k \alpha$ .

By the assumption, each  $TE_i \alpha \in \text{Im } E_i$ , so  $TE_i \alpha = E_i \beta_i$  for some  $\beta_i \in V$ .

Therefore,

$$E_i T \alpha = E_i (TE_1 \alpha + \dots + TE_k \alpha) = E_i (E_i \beta_1 + \dots + E_k \beta_k) = E_i E_i \beta_i = E_i \beta_i = TE_i \alpha.$$

So converse. Conversely, let  $\alpha \in \text{Im } E_i$ . Then,  $T\alpha = TE_i \alpha = E_i T \alpha \in \text{Im } E_i$ , so proved.  $\square$